1. Literature Review
The review covers prior suicidal ideation detection work spanning feature-engineering approaches (Shah et al.'s genetic algorithm + Random Forest), multitask learning (Benton et al., Ghosh et al.'s CEASE emotion/depression detection), and deep learning models (LSTM-CNN with word2vec, CNN-BiLSTM). It also covers context-modeling approaches (CRF-based, attention-based relation networks) and transformer advances, including domain-specific pretrained models (MentalBERT/MentalRoBERTa) and time-aware models (STATENet) — establishing performance benchmarks the current paper aims to beat.

2. Methodology
The authors built three distinct deep learning architectures, one tailored to each of three benchmark datasets (SuicideDetection, CEASEv2.0, SWMH). After dataset-specific preprocessing (contraction expansion, stopword removal except "I", spell correction, lemmatization as needed), text was converted to vectors using GloVe, FastText, or transformer-based sentence embeddings, then fed into BiLSTM/2BiLSTM networks (for the first two datasets) or a BERT transformer (for SWMH), with dropout, batch normalization, and hyperparameter tuning (via RMSprop/Adam/Nadam optimizers). Models were evaluated using accuracy, precision, recall, F1-score, and AUC.

3. Limitations
Key limitations: (a) self-reported social media data may not reflect true feelings (sarcasm, quotations, and slang cause misclassification); (b) privacy/ethical constraints limit access to demographic data (age, gender, socioeconomic status) that could improve or contextualize predictions; (c) the temporal nature of posts makes real-time risk detection difficult, as posts may appear only hours before an attempt; (d) binary classification oversimplifies suicide risk into 0/1 labels without capturing severity/degree; (e) potential demographic/linguistic biases inherited from training data could cause false negatives in underrepresented groups — a serious concern since missed true-risk cases (false negatives) are far more dangerous than false positives. These could be addressed by testing across diverse demographic/linguistic datasets, incorporating human-in-the-loop review, exploring multimodal data (text + visual/audio), and building severity-aware (non-binary) classification.

4. Method Used
Three separate deep neural network architectures: (1) **FastText + 2-layer BiLSTM** for the SuicideDetection dataset (binary classification), (2) **FastText + BiLSTM with initial bias correction** for CEASEv2.0 (binary classification), and (3) **BERT transformer** for the SWMH dataset (5-class multi-class classification).

Deep Dive

5. Advantages and Disadvantages

Advantages:

- Achieved state-of-the-art performance on two of three datasets (SuicideDetection: 0.97 F1 vs. prior best 0.95 accuracy; CEASEv2.0: 0.75 F1 vs. prior best 0.7435 F1), demonstrating the effectiveness of tailoring architecture to dataset characteristics rather than using one-size-fits-all models.
- The dataset-specific design philosophy (different embeddings, network depths, and preprocessing per dataset) is empirically justified — e.g., FastText's subword-level handling proved more robust and semantically stable (Table 9) than Word2Vec for out-of-vocabulary/informal social media text.
- Using dropout, batch normalization, and callbacks (early stopping, LR reduction) helped mitigate overfitting, particularly important given the very different dataset sizes (SuicideDetection is 46.4k times larger than CEASEv2.0).
- The BERT-based approach leverages self-attention to capture nuanced semantic relationships needed for fine-grained 5-class distinction (depression vs. anxiety vs. suicidewatch, etc.).

Disadvantages:

- The BERT model underperformed the domain-specific MentalRoBERTa (0.68 vs. 0.72 F1) on SWMH, showing that a general-purpose pretrained transformer without domain-specific fine-tuning leaves performance on the table.
- Models struggle with sarcasm, quotations, and nonsensical/slang text, as shown in the error analysis (e.g., "pretty good life" with high "happy" word frequency being misclassified as non-suicidal despite true suicidal intent).
- Shorter texts are disproportionately vulnerable to misclassification, since a few keywords carry outsized influence on the prediction (e.g., "tell," "story," "wanted" wrongly triggering a suicidal label).
- Binary classification cannot represent a spectrum of risk severity, an important clinical limitation acknowledged by the authors themselves.
- Heavy hyperparameter tuning (different dropout rates, batch sizes, epochs per dataset) reduces the "plug-and-play" generalizability of the specific architectures to new, unseen datasets.

6. How the Methods Work (Detailed)

General Pipeline (Fig. 1): Raw posts → preprocessing (HTML/link/special-character removal, contraction expansion, stopword removal except "I," dataset-specific steps like spell correction or lemmatization) → word/sentence embedding (Word2Vec, FastText, or transformer-based

sentence embeddings) → deep learning classifier (BiLSTM/Attention/Transformer with dropout and batch normalization) → binary (sigmoid) or multi-class (softmax) output.

SuicideDetection Model: Input text (up to 188 tokens) is embedded using **FastText** (300-dimensional, crawl-300d-2M-subword vectors), which — unlike Word2Vec — builds representations from subword character n-grams, letting it handle typos, informal spelling, and out-of-vocabulary words common on Reddit. The embedded sequence passes through a **dropout layer (0.3)**, then **two stacked BiLSTM layers (128 units each)**, where the first layer's return_sequences=True setting feeds its full hidden-state sequence into the second layer, allowing the model to capture both fine-grained token-level and higher-order sequence-level patterns in both forward and backward directions. This is followed by two dense layers (32 and 16 units, ReLU) with dropout (0.3) between them, a batch normalization layer, and a final single-unit sigmoid output for binary classification.

CEASEv2.0 Model: Given the dataset's small size (4,932 sentences) and class imbalance (64% non-depression, 36% depression), a shallower model was used: FastText embedding → single BiLSTM (32 units) → dense (8 units, ReLU) → dropout (0.25) → dense (4 units, ReLU) → dropout (0.25) → batch normalization → sigmoid output. Critically, a **bias initializer** was applied to the output layer to counteract the class imbalance, nudging initial predictions toward the majority class distribution to speed up convergence and avoid instability.

SWMH Model (BERT): Input text (256 tokens) is fed into a full **BERT-base transformer** (12 encoder blocks, 768-dimensional hidden states, ~108.7M parameters), which uses **multi-head self-attention** in each block to weigh relationships between all token pairs simultaneously, capturing contextual and semantic nuance far beyond word-level embeddings. BERT's pooled output passes through a dense layer (512 units) and then a final dense layer with **softmax activation** producing probabilities across the 5 classes (depression, suicidewatch, anxiety, offmychest, bipolar). Training used sparse categorical cross-entropy loss, since labels are integer-encoded rather than one-hot for this multi-class setup.

7. Suitable Contexts/Problems

These approaches are best suited for:

- Large-scale screening of publicly available social media text (Reddit, Twitter) where anonymity encourages more open self-disclosure than clinical settings — useful for population-level surveillance or as a flagging tool.
- Binary risk-flagging tasks (suicidal vs. non-suicidal) where FastText+BiLSTM combinations are computationally cheaper and perform very well given sufficiently large, relatively balanced training data (as with SuicideDetection's 232k balanced posts).
- Fine-grained differential classification tasks (distinguishing between related but distinct conditions — e.g., depression vs. anxiety vs. bipolar) where contextual, attention-based

transformer models like BERT are more suitable due to their superior capacity to capture subtle semantic differences.

- Small, imbalanced, and short-text datasets (like suicide notes) where shallower, well-regularized networks (fewer layers, bias correction) generalize better than deep architectures, which would overfit.
- It is less suited for: real-time crisis intervention (given the temporal lag/reactive nature of after-the-fact text analysis), highly sarcastic or culturally-specific text without domain adaptation, or contexts requiring severity/risk-level gradation rather than binary labels.

8. Main Results and Findings

- **SuicideDetection dataset**: FastText + 2BiLSTM model achieved **0.97 F1-score**, 0.9742 accuracy, and 0.996 AUC — outperforming the prior state-of-the-art CNN-BiLSTM+Word2Vec model (0.95 accuracy, validation loss 0.15 vs. this model's 0.08).
- **CEASEv2.0 dataset**: FastText + BiLSTM (with bias initializer) achieved **0.75 F1-score** and 0.70 AUC, outperforming the previous best GloVe+Bi-GRU multitask model (0.7435 F1).
- **SWMH dataset**: BERT achieved **0.68 F1-score**, outperforming the RN-based prior model (0.64 F1) but falling short of the domain-adapted MentalRoBERTa (0.72 F1) — a 4-point gap attributed to MentalRoBERTa's mental-health-specific pretraining.
- Comparative ML baselines (Tables 6–8) confirmed deep learning/transformer approaches generally outperformed classic ML (SVM reached 93.21% F1 on SuicideDetection vs. deep learning's 97%; various ML models on SWMH topped out around 66-67% F1 vs. BERT's 68%).
- FastText embeddings were shown to be more "stable" and semantically meaningful than Word2Vec across word-pair similarity tests (Table 9), helping explain its consistent selection across the two BiLSTM-based models.
- Error analysis revealed characteristic failure modes: short texts with suicide-associated keywords misclassified regardless of true sentiment, sarcastic/ambiguous phrasing, and quoted speech being misattributed to the author's own mental state.
- Class imbalance in SWMH was found to be moderate (ratio 0.408, within the "balanced" threshold used), while CEASEv2.0 was more clearly imbalanced (64%/36%), requiring bias-correction handling.

9. Overall Opinion and Critical Evaluation

This paper makes a methodologically thorough and practically grounded contribution by testing tailored architectures across three genuinely different data types (large balanced social media posts, small imbalanced suicide notes, and multi-class Reddit posts), rather than applying one

generic model everywhere — a design choice that is empirically justified and yields genuine state-of-the-art improvements on two datasets. The inclusion of an explicit error analysis, embedding similarity comparison (Table 9), and a dedicated ethics/bias discussion section (covering GDPR, demographic bias, and the asymmetric cost of false negatives vs. false positives) reflects unusually mature methodological and ethical rigor for this subfield, going beyond simply reporting benchmark numbers.

That said, several caveats limit the paper's real-world impact claims. The improvements over prior state-of-the-art are numerically modest (0.02 and 0.01 F1 points on two datasets) despite the added architectural complexity, raising the question of whether the gains are practically meaningful versus statistical noise from different train/test splits or hyperparameter search budgets — the paper does not report confidence intervals or statistical significance testing. The SWMH underperformance relative to MentalRoBERTa also undercuts the generalizability of a "one architecture per dataset" philosophy — it suggests domain-specific pretraining (as MentalRoBERTa uses) may matter more than architecture tuning alone, a point the authors acknowledge but don't further investigate (e.g., by fine-tuning their own BERT model on mental-health text before applying it to SWMH). Additionally, since all three datasets are Reddit- or note-based English text, the claimed real-world applicability (e.g., "a screening tool... used by mental health professionals or individuals") remains speculative without any clinical validation, cross-platform testing (Twitter/Facebook, as the authors themselves flag as future work), or non-English language testing. Finally, while the ethical discussion is commendable, the paper stops short of proposing concrete mitigation architecture (e.g., calibrated uncertainty estimates, fairness audits across demographic subgroups) beyond generally recommending human-in-the-loop oversight.